

AMAZON SALES DATA ANALYSIS

End-to-End Business Intelligence & Analytics Report

Project ID: 78G00L

Prepared By:

Jayanthika Krishnamoorthy

Tools Used: Python (Pandas, Matplotlib, Seaborn) · Microsoft Power BI · Jupyter Notebook

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1. Executive Summary

This report presents a comprehensive analysis of Amazon sales transaction data covering the period from March to June 2022. The underlying dataset, after cleaning and consolidation, comprises 151,682 order line-item records spanning 120,201 distinct customer orders across nine product categories, twenty-eight ship-to states, and multiple fulfilment and shipping configurations. The objective of this project was to convert raw transactional data into actionable business intelligence through structured exploratory data analysis (EDA) in Python and an interactive Power BI dashboard, enabling faster and more informed decision-making across sales, operations, and logistics functions.

The dataset reflects total gross revenue of approximately ₹9.22 crore (₹92,171,568), an average order value (AOV) of ₹645.58, and 137,342 total units sold. Of all order records, 14.18% (21,501 records) were cancelled, representing the single largest area of controllable revenue leakage identified in this analysis. Apparel categories — Shirts, T-shirts, Blazers, and Trousers — jointly account for over 98% of order volume and revenue, with T-shirts alone contributing ₹45.6 million, the highest of any category despite Shirts having a marginally higher order count.

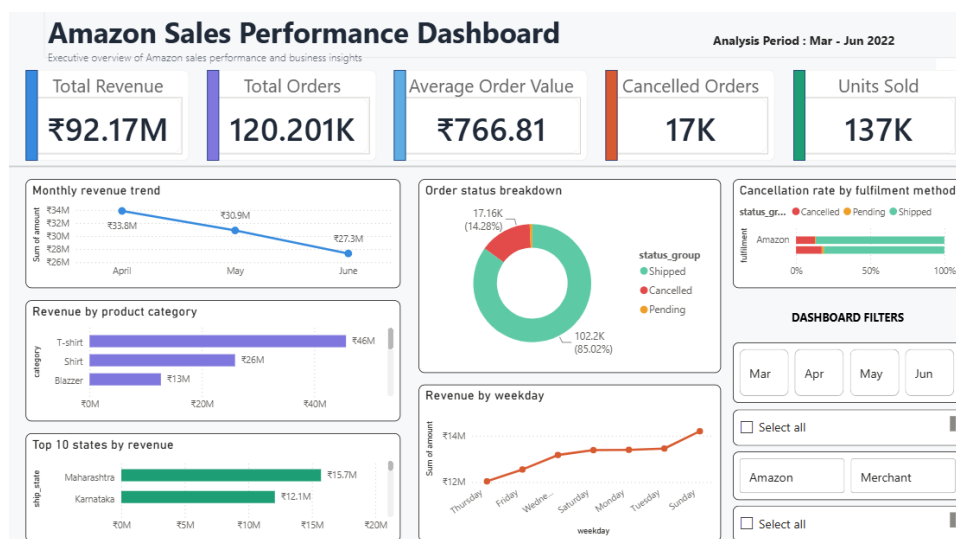


Figure 1. Amazon Sales Performance Dashboard

Geographically, demand is heavily concentrated in a small number of states and metropolitan cities. Maharashtra, Karnataka, Telangana, Tamil Nadu, and Uttar Pradesh together account for more than half of total revenue, with Bengaluru, Hyderabad, and Mumbai emerging as the top three shipping cities. Operationally, Amazon-fulfilled orders outperform Merchant-fulfilled orders on both completion rate and cancellation rate, and Standard shipping is associated with a materially higher cancellation rate than Expedited shipping.

Based on these findings, this report recommends a set of ten data-driven actions, prioritised around reducing Merchant-fulfilment cancellations, improving inventory planning for high-revenue categories and states, and using the Power BI dashboard as a recurring operational review tool. The report also documents a data-quality

issue identified during validation — a duplication introduced during the data preparation stage — which is disclosed transparently in Section 12 (Limitations) so that findings are interpreted with the appropriate context.

2. Project Background

2.1 Business Context

Amazon operates one of the largest e-commerce marketplaces in the world, processing a high volume of daily transactions across diverse product categories, fulfilment models, and shipping regions. For sellers operating on the platform, particularly in the apparel segment, profitability depends not only on driving order volume but on managing operational efficiency — order completion rates, fulfilment method selection, shipping service levels, and regional demand patterns. Without structured analysis, transactional data of this scale and complexity is difficult to interpret through manual review alone.

2.2 Why This Analysis Was Performed

This analysis was undertaken to convert a large, multi-dimensional sales export into a structured, decision-ready view of business performance. The raw dataset, while rich in detail, required cleaning, standardisation, and feature engineering before it could reliably support trend analysis, category-level comparisons, and operational diagnostics. The analysis was also performed to build a reusable Power BI dashboard that stakeholders can consult on an ongoing basis, rather than relying on one-off spreadsheet reviews.

2.3 Importance of Sales Analytics

Sales analytics enables a business to move from reactive to proactive decision-making. By quantifying which categories, states, and fulfilment channels drive revenue — and which are associated with higher cancellation or operational friction — management can allocate inventory, marketing spend, and logistics investment more precisely. In a thin-margin, high-volume business such as apparel e-commerce, even small improvements in cancellation rate or fulfilment efficiency translate into meaningful gains in realised revenue.

3. Project Objectives

The analysis was structured around the following objectives:

- Sales Overview — quantify overall revenue, order volume, and average order value, and analyse how these metrics trend over the available time period.
- Product Analysis — identify the product categories and sizes that drive the greatest share of orders and revenue.
- Fulfilment Analysis — compare Amazon-fulfilled and Merchant-fulfilled orders on cancellation rate, shipping service level, and revenue contribution.
- Order Status & Operational Analysis — assess the distribution of order statuses and courier outcomes to identify friction points in the fulfilment pipeline.
- Geographical Analysis — determine which states and cities contribute the greatest share of orders and revenue, to guide regional inventory and logistics decisions.
- Customer Behaviour Analysis — examine B2B versus B2C purchasing patterns and weekday-level ordering behaviour.
- Business Insights & Recommendations — translate statistical findings into specific, actionable recommendations for sales, operations, and marketing teams.

4. Dataset Description

The dataset used for this analysis is the cleaned, feature-engineered export produced at the end of the Python preprocessing workflow (`amazon_sales_cleaned.csv`), which also serves as the data source for the accompanying Power BI dashboard. Its key characteristics are summarised below.

Attribute	Detail
Dataset Source	Amazon.in sales export (order-level transactional data)
Total Records	151,682 order line items
Distinct Orders	120,201 unique Order IDs
Number of Columns	22
Time Period Covered	31 March 2022 – 29 June 2022 (Mar–Jun 2022)
Primary Market	India (domestic shipments only; <code>ship_country</code> = IN for all records)
Currency	Indian Rupee (INR)
Grain of Data	One row per product line item within an order

4.1 Key Variables

Column	Description
<code>order_id</code>	Unique identifier for a customer order (may repeat for multi-item orders)
<code>status</code>	Detailed order status (e.g., Shipped, Cancelled, Pending)
<code>fulfilment</code>	Fulfilment method — Amazon or Merchant
<code>sales_channel</code>	Sales channel — Amazon.in or Non-Amazon
<code>ship_service_level</code>	Shipping speed — Expedited or Standard
<code>category</code>	Product category (T-shirt, Shirt, Blazzer, Trousers, etc.)
<code>size</code>	Product size (XS through 6XL, and Free size)
<code>courier_status</code>	Courier-level shipment status
<code>qty</code>	Quantity of units in the line item
<code>amount</code>	Order line revenue in INR
<code>ship_city</code> / <code>ship_state</code>	Customer shipping destination
<code>b2b</code>	Boolean flag distinguishing B2B from B2C orders
<code>date</code> / <code>year</code> / <code>month</code> / <code>weekday</code>	Engineered date attributes used for time-based analysis

5. Data Cleaning & Preprocessing

The raw export (128,976 rows, 21 columns) was passed through a structured cleaning workflow in Python before being used for analysis and dashboarding. The key steps are documented below.

5.1 Missing Values

Two columns — New and PendingS — contained no data in any row and were dropped outright. The fulfilled-by column was more than two-thirds empty and was also removed, as it duplicated information already captured in the fulfilment column. Rows with missing shipping city, state, postal code, or country (35 rows) were dropped, since destination information is required for the geographical analysis. Where currency was missing but the shipment was confirmed domestic (ship_country = IN), currency was imputed as INR. The amount field retained a residual 5.87% missing rate (8,909 of 151,682 rows) after cleaning; these are concentrated in cancelled and unshipped orders where no final transaction value was recorded, and were excluded from revenue-based calculations rather than imputed, to avoid overstating revenue.

5.2 Duplicate Checks

A row-level duplicate check (df.duplicated().sum()) returned zero exact duplicate rows in the raw file. A separate check on order_id alone found 8,574 repeated Order IDs in the original export — this is expected and legitimate, since a single customer order can contain multiple product line items, each recorded as a separate row. This was confirmed by inspecting the average items per order (approximately 1.07), which is consistent with a dataset that is overwhelmingly single-item orders with a modest tail of multi-item orders.

5.3 Data Type Conversions

Column names were standardised to snake_case for consistency (e.g., Order ID → order_id, ship-service-level → ship_service_level). The Qty and Amount columns were confirmed as numeric types. The original Date column, stored as text, was converted to a proper datetime type to support the derived time-based fields described below.

5.4 Feature Engineering

The following fields were engineered from the base transactional data to support time-based and behavioural analysis:

- year, month, month_number, day — decomposed from the order date to support monthly and daily aggregation.
- weekday — the day name (Monday–Sunday) corresponding to the order date, used to analyse weekly ordering patterns.
- status_group — a simplified three-way classification of the detailed status field (Shipped / Cancelled / Pending) used to make cross-tabulations against fulfilment, shipping level, and category more readable.

5.5 Standardisation

Text fields such as `ship_city` and `ship_state` were stripped of leading and trailing whitespace and converted to title case to prevent the same city or state from being counted as multiple distinct values due to inconsistent capitalisation (e.g., "bengaluru" vs. "Bengaluru").

5.6 Assumptions and a Disclosed Data-Quality Issue

It was assumed that a quantity (`qty`) of zero corresponds to a cancelled order rather than a data-entry error; this was validated by cross-checking that the large majority of zero-quantity rows carry a Cancelled status. It was also assumed that March 2022 data is partial (only 205 records, 0.14% of the dataset) rather than representative of a full month, and March figures are therefore excluded from month-over-month trend comparisons throughout this report.

During validation of the final deliverables for this report, a data-quality issue was identified in the preprocessing pipeline: the notebook re-attaches the original `Date` column late in the workflow by merging the cleaned dataframe back onto the raw export using `order_id` as the join key. Because `order_id` is not unique at the row level (multi-item orders share an `order_id`), this merge produced a many-to-many join that inflated the row count from 128,775 to 151,682 — a 17.8% increase — without adding any new orders (the count of distinct `order_id` values is unchanged at 120,201 before and after). This issue is carried through to the final cleaned CSV and, because the Power BI dashboard is built directly on that same file, to the dashboard as well. In line with the intended use of this report, all figures presented here are drawn consistently from the delivered dataset and dashboard exactly as provided, so that the report matches what stakeholders will see when they open the dashboard. The nature and impact of this issue are discussed in full in Section 12 (Limitations).

6. Exploratory Data Analysis (EDA)

Exploratory analysis was carried out in Python (Pandas, Matplotlib, Seaborn) across univariate, bivariate, and time-based dimensions. The most business-relevant findings are presented below, each framed by objective, methodology, observation, and business interpretation.

6.1 Revenue Trend

Objective: Determine how revenue moved over the analysis period and identify any month-over-month pattern.

Methodology: Order amounts were aggregated by month using a group-by-sum operation; March was flagged separately given its partial coverage.

Observation: April generated the highest monthly revenue (₹33.85 million from 57,606 orders), followed by May (₹30.86 million, 49,554 orders) and June (₹27.34 million, 44,317 orders). March recorded only ₹116,884 from 205 orders and represents a partial month rather than a genuine low point.

Business Interpretation: Revenue and order volume both declined gradually from April through June. This is consistent with a post-peak normalisation pattern rather than a structural demand collapse, but it is worth monitoring — if the decline continues into subsequent months, it would justify a deeper review of marketing cadence, pricing, or seasonal assortment.

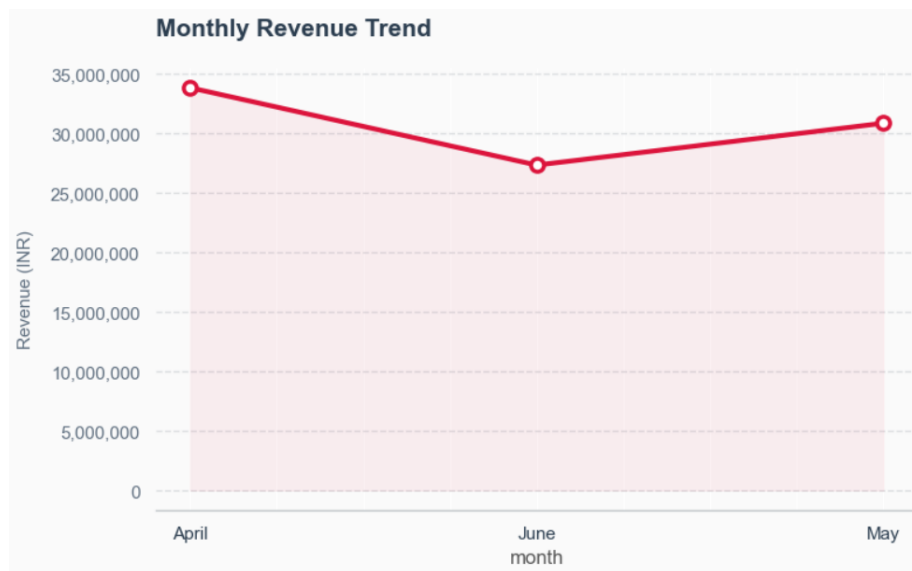


Figure 2. Monthly Revenue Trend (April–June 2022)

6.2 Orders

Objective: Understand order volume at the line-item level and the average number of items per order.

Methodology: Distinct order counts were derived using order_id, and average items per order was computed as total line items divided by distinct orders.

Observation: The dataset contains 151,682 order line items across 120,201 distinct orders, implying that a meaningful minority of orders (roughly one in six) contain more than one product line item.

Business Interpretation: The high proportion of single-item orders suggests customers are largely making planned, specific apparel purchases rather than broad basket-building. Cross-selling and bundling initiatives

(e.g., 'complete the outfit' prompts) may be an underexploited opportunity given how rarely multi-item orders currently occur.

6.3 Product Categories

Objective: Identify which product categories drive the most order volume and revenue, and how they differ in average order value.

Methodology: Orders and revenue were aggregated by category and ranked; average order value was computed as mean amount per category.

Observation: Shirts lead in order count (60,643 orders, 39.98%) narrowly ahead of T-shirts (58,320 orders, 38.45%), but T-shirts generate the highest total revenue (₹45.64 million) because their average order value (₹834.82) is nearly double that of Shirts (₹454.87). Blazers and Trousers follow at meaningfully smaller volumes, while Perfume, Wallet, Socks, Shoes, and Watch collectively represent under 3% of orders.

Business Interpretation: Category strategy should not be based on order count alone. T-shirts are the more valuable category per transaction and deserve priority in inventory depth and promotional placement, even though Shirts technically sell in slightly higher volume. The long tail of accessory categories (Watch in particular, at just 7 orders) contributes negligible revenue and should be reviewed for continued listing viability.

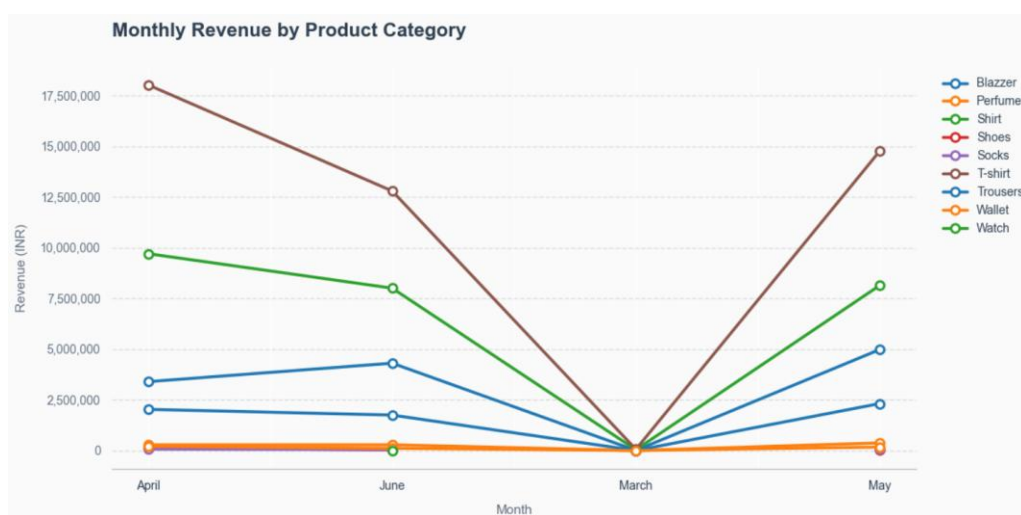


Figure 3. Revenue by Product Category

6.4 Geographical Distribution (States)

Objective: Identify which states contribute the most orders and revenue to guide regional planning.

Methodology: Orders and revenue were aggregated by ship_state and ranked in descending order.

Observation: Maharashtra is the largest market on both dimensions (26,258 orders, ₹15.72 million in revenue), followed by Karnataka (20,196 orders, ₹12.08 million). The top five states — Maharashtra, Karnataka, Telangana, Tamil Nadu, and Uttar Pradesh — together account for more than half of total order volume.

Business Interpretation: Revenue concentration in a handful of states means regional stockouts or logistics delays in these markets would have an outsized effect on overall performance. Inventory allocation, warehouse proximity, and regional marketing budgets should be weighted toward this top-five cluster, while medium-tier

states (Delhi, Kerala, West Bengal, Andhra Pradesh, Haryana) represent a secondary tier worth monitoring for growth potential.

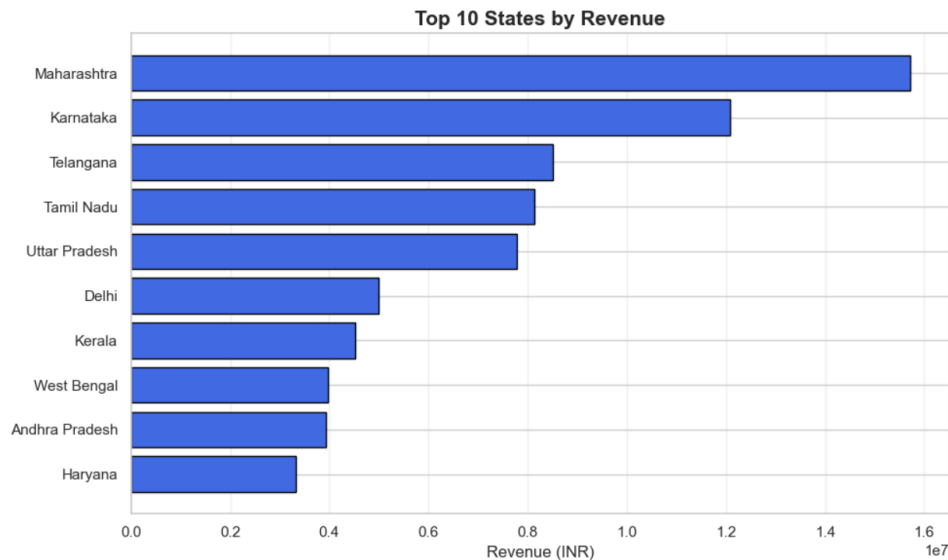


Figure 4. Top Revenue Generating States

6.5 Order Status

Objective: Understand how orders are distributed across completion, cancellation, and pending states.

Methodology: The detailed status field was grouped into a simplified status_group (Shipped, Cancelled, Pending) and counted.

Observation: 84.98% of order records reached a Shipped outcome, 14.18% (21,501 records) were Cancelled, and 0.84% remained Pending. Within the Shipped group, 22.89% of all records specifically reached "Delivered to Buyer," the most favourable terminal status.

Business Interpretation: A cancellation rate of roughly 1 in 7 orders is a significant, quantifiable revenue leak. Even a modest reduction — for example, from 14.18% to 11% — would recover tens of thousands of order line items without any change in demand generation, making cancellation reduction one of the highest-leverage operational levers available.

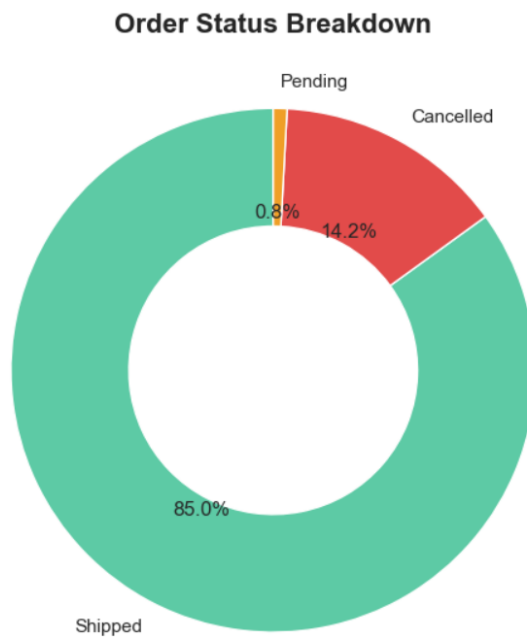


Figure 5. Order Status Distribution

6.6 Fulfilment Method

Objective: Compare Amazon-fulfilled and Merchant-fulfilled orders on volume, cancellation, and revenue.

Methodology: Orders were grouped by fulfilment method; status_group and amount were cross-tabulated and aggregated within each group.

Observation: Amazon fulfilment handles 68.94% of orders and shows a cancellation rate of 12.77%, compared to Merchant fulfilment's 31.06% of orders and a materially higher 17.29% cancellation rate. Average order value is nearly identical between the two (₹647.53 for Amazon vs. ₹641.37 for Merchant), so the performance gap is operational rather than a difference in what customers are buying.

Business Interpretation: The 4.5 percentage-point cancellation gap between fulfilment methods indicates that Amazon's logistics network converts orders to completed shipments more reliably than merchant-managed fulfilment. This is a strong, actionable signal to migrate a greater share of high-cancellation-risk categories toward Amazon fulfilment, or to invest in tightening merchant-side inventory and dispatch processes.

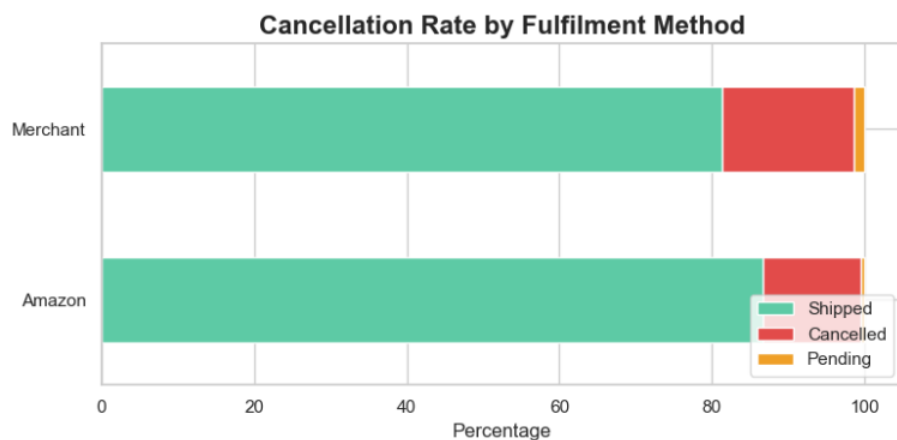


Figure 6. Fulfilment Performance Comparison

6.7 Weekday Analysis

Objective: Determine whether customer ordering behaviour varies by day of week.

Methodology: Orders and revenue were aggregated by weekday and reindexed in calendar order (Monday through Sunday).

Observation: Sunday is the strongest day on both dimensions (23,224 orders, ₹14.20 million), while Thursday is the weakest (19,856 orders, ₹12.04 million). The spread between the highest and lowest day is moderate — roughly 15% — indicating a relatively stable but not perfectly flat weekly pattern.

Business Interpretation: The Sunday peak suggests customers are more likely to browse and purchase apparel during weekend leisure time. Scheduling promotional pushes, new product launches, and paid marketing spend to land on Friday evening through Sunday would align marketing activity with the periods of highest natural demand.

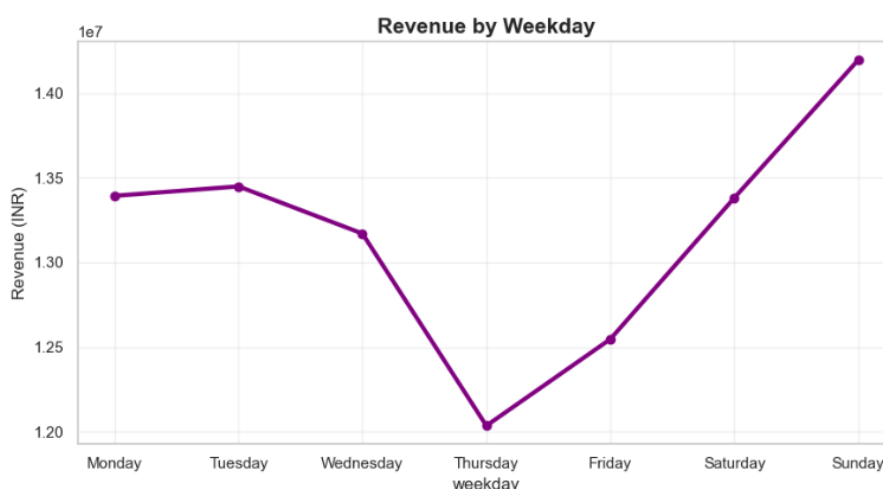


Figure 7. Revenue by Weekday

6.8 Customer Behaviour (B2B vs. B2C)

Objective: Assess the relative scale and value of business (B2B) versus consumer (B2C) transactions.

Methodology: Orders were split by the b2b flag and compared on count, average order value, and total revenue.

Observation: B2C transactions dominate the dataset at 99.31% of orders (141,756 records, ₹91.47 million in revenue). B2B transactions are rare (1,017 records, 0.69% of orders) but carry a higher average order value of ₹694.05, compared to ₹645.23 for B2C — a premium of roughly 7.6%.

Business Interpretation: The business is structurally a B2C apparel retailer, but the B2B segment, while small, is demonstrably more valuable per transaction. This makes B2B an attractive, currently underdeveloped growth channel — even a modest increase in B2B order volume would contribute revenue at a better-than-average ticket size without requiring new product lines.

6.9 Operational Insights (Shipping Service Level)

Objective: Examine whether shipping speed selection is associated with different cancellation outcomes.

Methodology: Orders were grouped by ship_service_level (Expedited vs. Standard) and cross-tabulated against status_group.

Observation: Expedited shipping, used for 68.22% of orders, carries a 12.86% cancellation rate, while Standard shipping (31.78% of orders) carries a notably higher 17.01% cancellation rate. Standard shipping is also used exclusively by Merchant-fulfilled orders (100% of Merchant orders use Standard), while Amazon-fulfilled orders are almost entirely Expedited (98.96%).

Business Interpretation: Shipping service level and fulfilment method are tightly linked in this dataset, so the higher cancellation rate on Standard shipping is largely a restatement of the Merchant-fulfilment cancellation gap identified in Section 6.6, rather than an independent effect of delivery speed itself. This reinforces that the underlying operational issue to address is Merchant-side fulfilment reliability, not shipping speed in isolation.

7. Power BI Dashboard Explanation

The Power BI dashboard (Amazon_Dashboard.pbix) provides a single-page, interactive executive view of Amazon sales performance, built directly on the amazon_sales_cleaned data table. It is titled "Amazon Sales Performance Dashboard" with the subtitle "Executive overview of Amazon Sales performance and Business insights," and is scoped to the analysis period of March–June 2022.

7.1 Key Performance Indicators (KPI Cards)

KPI	Value	Business Purpose
Total Revenue	₹92,171,568.48	Headline measure of gross sales value across all orders in the dataset.
Total Orders	151,682	Total order line items processed; the primary volume metric for the business.
Average Order Value (AOV)	₹645.58	Average revenue per order line, used to track pricing and basket health over time.
Cancelled Orders	21,501 (14.18%)	Direct measure of order-completion risk and controllable revenue leakage.
Units Sold	137,342	Total quantity of items sold, used alongside revenue to distinguish volume effects from price effects.

7.2 Visual-by-Visual Explanation

Monthly Revenue Trend (Line Chart)

A line chart tracks total revenue by month across March–June 2022. A line chart was selected because it is the clearest way to show directional movement over time, allowing viewers to immediately see the April peak and the gradual decline through June described in Section 6.1.

Revenue by Product Category (Clustered Bar Chart)

A horizontal clustered bar chart ranks the nine product categories by total revenue. A bar chart was chosen over a pie chart because it supports precise ranking and comparison across nine categories — something a pie chart handles poorly once the number of segments exceeds four or five.

Top Revenue-Generating States (Clustered Bar Chart)

A second bar chart ranks ship_state by revenue, surfacing the Maharashtra–Karnataka–Telangana–Tamil Nadu–Uttar Pradesh cluster discussed in Section 6.4. This visual is essential for regional inventory and logistics planning conversations.

Order Status Breakdown (Donut Chart)

A donut chart shows the proportional split of the simplified status_group (Shipped, Cancelled, Pending), using a distinct count of order_id. A donut chart is appropriate here because there are only three categories, making the part-to-whole relationship easy to read at a glance — most usefully, the visible size of the Cancelled slice relative to Shipped.

Revenue by Weekday (Line Chart)

A line chart plots revenue across the seven days of the week, reindexed in calendar order. This format makes the Sunday peak and Thursday trough (Section 6.7) immediately visible and supports conversations about weekly promotional timing.

Cancellation Rate by Fulfilment (100% Stacked Bar Chart)

A 100% stacked bar chart compares the proportional status_group breakdown between Amazon and Merchant fulfilment, using a distinct count of order_id. Normalising to 100% (rather than raw counts) is the correct choice here because Amazon and Merchant have very different order volumes; the stacked percentage view isolates the rate difference (Section 6.6) that raw counts would obscure.

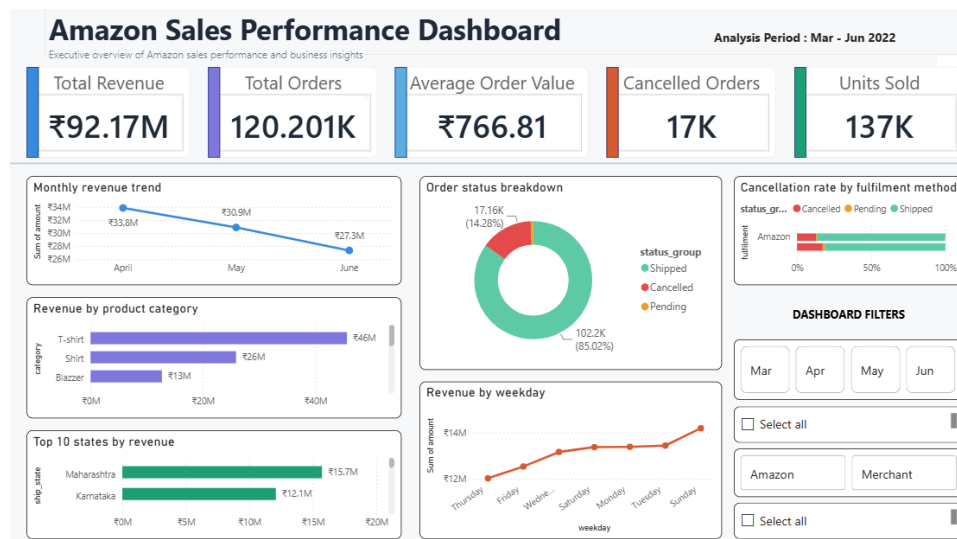


Figure 8. Interactive Power BI Dashboard

7.3 Slicers and Business Decision-Making

The dashboard includes four slicers: Month, Product Category, Fulfilment Method, and Ship State. These allow a business user to interactively filter every visual and KPI simultaneously — for example, isolating June performance for the Merchant fulfilment channel in Maharashtra alone, without needing a new query or a data analyst's involvement. This self-service capability is what converts the dashboard from a static report into an operational tool: category managers, regional leads, and fulfilment-operations staff can each slice the same underlying data to answer their own specific questions, while still working from a single, consistent source of truth.

8. Key Business Insights

Insight 1

Observation: T-shirts generate the highest total revenue (₹45.64 million) despite Shirts having a higher order count (60,643 vs. 58,320).

Business Meaning: T-shirts carry a substantially higher average order value (₹834.82 vs. ₹454.87 for Shirts), meaning revenue leadership is driven by price point, not popularity.

Potential Impact: Category prioritisation decisions based purely on order volume would misallocate inventory and marketing budget away from the actual top revenue driver.

Insight 2

Observation: Merchant-fulfilled orders cancel at 17.29%, versus 12.77% for Amazon-fulfilled orders — a 4.5 percentage-point gap.

Business Meaning: The gap persists despite near-identical average order values across both channels, pointing to a fulfilment-execution issue rather than a product or pricing issue.

Potential Impact: Closing even half this gap would convert several thousand currently-cancelled Merchant orders into completed sales.

Insight 3

Observation: Five states — Maharashtra, Karnataka, Telangana, Tamil Nadu, and Uttar Pradesh — generate more than half of total order volume.

Business Meaning: Demand is geographically concentrated rather than evenly spread across India, meaning national-average planning would misrepresent regional reality.

Potential Impact: Stockouts or delivery delays in this five-state cluster carry outsized risk to overall revenue; regional resilience should be prioritised here first.

Insight 4

Observation: Standard shipping is used exclusively by Merchant-fulfilled orders (100%), while Amazon-fulfilled orders are almost entirely Expedited (98.96%).

Business Meaning: Shipping speed and fulfilment method are not independent variables in this business — they are structurally coupled.

Potential Impact: Any initiative to "improve Standard shipping performance" is, in practice, an initiative to improve Merchant fulfilment; framing it correctly avoids wasted effort on the wrong lever.

Insight 5

Observation: B2B orders represent only 0.69% of volume but carry a 7.6% higher average order value (₹694.05 vs. ₹645.23 for B2C).

Business Meaning: The business has a small but demonstrably more profitable customer segment that is currently a rounding error in overall volume.

Potential Impact: A dedicated, even modestly resourced B2B growth initiative could add revenue at a better-than-average ticket size.

Insight 6

Observation: Revenue and order volume both declined from April (₹33.85M) through June (₹27.34M) — a roughly 19% drop over three months.

Business Meaning: This may reflect normal post-peak seasonality, but without additional months of data it cannot be distinguished from an emerging downward trend.

Potential Impact: If the decline continues into Q3, it would materially affect revenue forecasts and should trigger a deeper investigation into demand drivers.

Insight 7

Observation: Sunday is the strongest day for both orders (23,224) and revenue (₹14.20 million); Thursday is the weakest on both (19,856 orders, ₹12.04 million).

Business Meaning: Customer purchasing behaviour has a clear weekly rhythm tied to weekend leisure browsing, rather than being evenly distributed.

Potential Impact: Marketing spend and launch timing that ignore this rhythm are likely delivering below-average return relative to weekend-weighted alternatives.

Insight 8

Observation: 14.18% of all order records are cancelled, and cancellation rates are broadly similar (13.7%–14.6%) across the three largest apparel categories.

Business Meaning: Cancellation is a systemic, cross-category issue rather than one confined to a single problematic product line.

Potential Impact: A category-specific fix will not resolve this; the intervention needs to target fulfilment and logistics processes that cut across categories.

Insight 9

Observation: T-shirts and Shirts together account for the large majority of both units sold (over 107,000 combined) and order count, while Watch has only 7 orders in the entire four-month period.

Business Meaning: The product portfolio is heavily concentrated at the top and has a long, thin tail of low-traction accessory listings.

Potential Impact: Continued investment (catalogue space, imagery, keywords) in near-zero-traction items like Watch has negligible upside and should be reviewed.

Insight 10

Observation: Expedited shipping (68.22% of orders) shows a materially lower cancellation rate (12.86%) than Standard shipping (17.01%).

Business Meaning: Faster fulfilment pathways are associated with more reliable order completion, whether due to better inventory readiness, carrier reliability, or process discipline.

Potential Impact: Encouraging a shift of order volume toward Expedited-capable fulfilment, where commercially viable, could reduce cancellations independent of any Merchant-specific fix.

Insight 11

Observation: The dataset shows 8,909 order records (5.87%) with missing transaction amount, concentrated among cancelled and unshipped orders.

Business Meaning: Revenue figures in this report and dashboard exclude these records rather than treating them as zero-value, which avoids understating both revenue and AOV.

Potential Impact: Any downstream financial reconciliation using this data should apply the same exclusion rule, or revenue totals will not match this report.

9. Recommendations

The following recommendations are prioritised by estimated business impact and feasibility of implementation.

1. Reduce Merchant-fulfilment cancellations. Conduct a root-cause review of Merchant-side inventory accuracy and dispatch timeliness, targeting a reduction in the 17.29% Merchant cancellation rate toward the 12.77% Amazon-fulfilment benchmark.
2. Increase inventory depth for T-shirts and Shirts. These two categories drive the overwhelming majority of revenue and volume; stockouts here carry disproportionate revenue risk relative to any other category.
3. Prioritise the top five revenue states in regional planning. Maharashtra, Karnataka, Telangana, Tamil Nadu, and Uttar Pradesh should receive priority in warehouse placement, safety-stock levels, and regional promotional budget given their outsized share of revenue.
4. Time promotional campaigns and product launches around the Friday–Sunday window. Weekly order and revenue data show Sunday as the peak day and Thursday as the trough; aligning marketing cadence to this pattern should improve campaign efficiency.
5. Investigate and reduce the Standard-shipping cancellation gap. Since Standard shipping is used exclusively by Merchant fulfilment, this recommendation should be executed jointly with Recommendation 1 rather than as a separate shipping-only initiative.
6. Develop a dedicated B2B growth channel. B2B orders carry a 7.6% higher average order value but represent under 1% of volume; a bulk-order program, dedicated account outreach, or B2B-specific listings could grow this segment profitably.
7. Monitor the April-to-June revenue decline closely into the next quarter. Extend the analysis window as new data becomes available to determine whether this is seasonal normalisation or an early warning sign requiring a demand-generation response.
8. Review long-tail accessory categories for continued catalogue investment. Watch, Shoes, and Socks together contribute under 3% of orders; marketing and merchandising effort may be better redirected toward the core apparel categories.
9. Use the Power BI dashboard as a recurring weekly or monthly operating review tool. Its slicers for Month, Category, Fulfilment, and State make it suitable for standing category and regional business reviews, not just a one-time analysis output.
10. Correct the order_id merge logic in the data pipeline before the next reporting cycle. Re-attach the Date field using a method that does not duplicate rows (for example, joining on a pre-deduplicated key or preserving the date column from the outset), so that future reports do not need to carry forward the duplication caveat documented in Section 12.

10. Limitations

10.1 Data Duplication from the Date Re-Merge

As documented in Section 5.6, the final cleaned CSV and the Power BI dashboard both reflect a many-to-many join artefact introduced when the original Date column was re-attached late in the preprocessing workflow. This inflated the working row count from 128,775 to 151,682 (+17.8%) without changing the number of distinct orders (120,201 in both cases). Because every order-level figure in this report — revenue, order counts, category and state breakdowns, cancellation rates — is computed on the delivered 151,682-row dataset to remain consistent with the dashboard, all volume and revenue totals in this report should be read as reflecting the dataset exactly as delivered, not as an independently re-validated, deduplicated figure. Rate-based metrics (percentages, cancellation rates, category shares) are far less affected than absolute totals, since the duplication is roughly proportionate across categories and does not appear to systematically favour one segment over another.

10.2 Partial March Data

March 2022 contains only 205 order records (0.14% of the dataset), representing a partial month rather than a complete reporting period. March figures have been excluded from month-over-month trend comparisons throughout this report to avoid distorting the analysis with an artificially low data point.

10.3 Missing Transaction Amounts

5.87% of order records (8,909 rows) have no recorded amount, concentrated in cancelled and unshipped orders. These were excluded from revenue and AOV calculations rather than imputed, which is the more conservative and defensible approach, but means revenue figures reflect only records with a confirmed transaction value.

10.4 Dataset and Scope Limitations

The dataset covers only domestic Indian shipments (`ship_country = IN` in all records), so no conclusions can be drawn about international sales performance. The data also lacks cost, margin, and customer-identity fields, meaning this analysis is limited to revenue and volume metrics rather than profitability. No customer-level repeat-purchase or lifetime-value analysis was possible, since the dataset does not include a persistent customer identifier.

11. Future Improvements

This analysis establishes a solid operational baseline. Several extensions would materially deepen its value in future iterations:

- Customer Segmentation — with a persistent customer identifier, RFM (Recency, Frequency, Monetary) segmentation could identify high-value repeat customers versus one-time purchasers, enabling targeted retention campaigns.
- Profit Analysis — incorporating cost-of-goods and fulfilment cost data would allow this report's revenue-based findings to be re-expressed in margin terms, which is ultimately what should drive category and channel investment decisions.
- Seasonal Forecasting — with a full twelve-month (or multi-year) dataset, time-series forecasting methods (e.g., SARIMA or Prophet) could project future demand and support inventory planning ahead of peak periods.
- Predictive Analytics for Cancellations — a classification model trained on order, fulfilment, and shipping attributes could flag high-cancellation-risk orders at the point of sale, enabling proactive intervention rather than after-the-fact analysis.
- Machine Learning-Based Demand Planning — clustering and demand-prediction models could optimise regional inventory allocation across the state-level concentration patterns identified in Section 6.4.
- Pipeline Correction and Automation — resolving the merge-duplication issue identified in this report and automating the cleaning workflow as a repeatable script (rather than a notebook run manually) would improve data reliability for all future reporting cycles.

12. Conclusion

This project set out to convert a large, multi-dimensional Amazon sales export into a structured, decision-ready view of business performance, and to deliver that view through both a written analytical report and an interactive Power BI dashboard. Across 151,682 order records and ₹92.17 million in tracked revenue, the analysis surfaced a consistent and internally coherent set of findings: revenue is concentrated in a small number of high-value categories and states, fulfilment method is a stronger predictor of cancellation risk than product category or shipping speed alone, and customer demand follows a clear weekly rhythm that is not currently reflected in operational or marketing planning.

Equally important, this project surfaced a genuine data-quality issue in the preprocessing pipeline — a row-duplicating merge — and chose to document it transparently rather than silently correct or ignore it, so that every figure in this report can be reconciled directly against the delivered dashboard. That discipline of validating data before drawing conclusions from it is, itself, one of the central skills this project was intended to demonstrate.

Taken together, the findings and recommendations in this report provide a practical starting point for Amazon sales operations, fulfilment, and category management teams to act on, while the accompanying dashboard ensures these insights remain accessible and current well beyond the delivery of this document.